



Introduction to Cyber Security and Ethical Hacking

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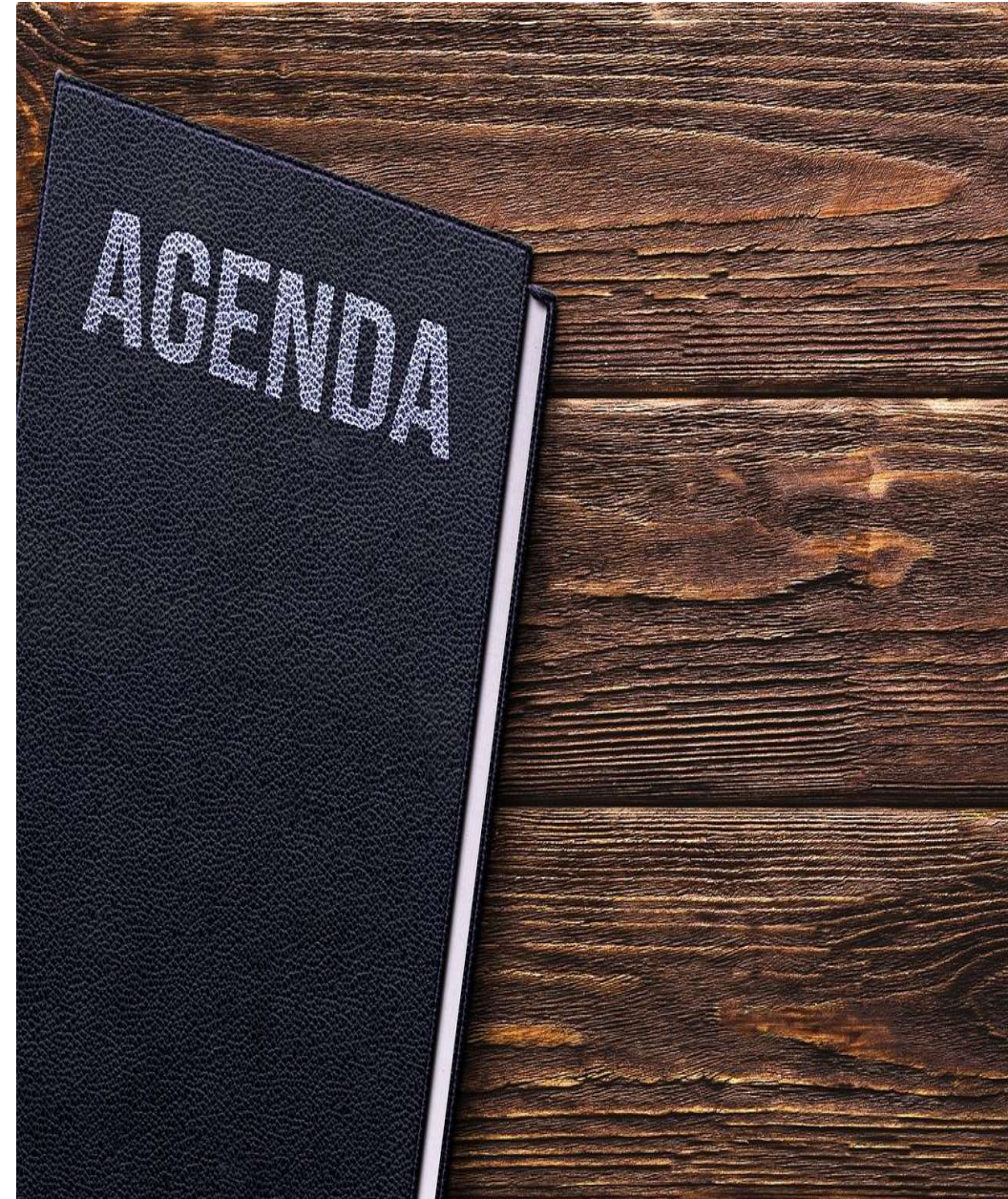
Scientist E, RISE

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- Introduction to Cyber Security
 - CIA, Cyber Security
- Introduction to Ethical Hacking
 - Types of Hacker Classes
 - Red team , Blue Team ,Grey Team
 - Ethical Hackers and Crackers
 - Goals of Attackers
 - Security, Functionality and Ease of Use Triangle
 - Defining the Skills Required to become an Ethical Hacker
 - Steps of Ethical Hacking
- Introduction to Kali Linux

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What is Security ?

▶ Security according to two boys of 10 years old.



Security ?

- ▶ Security According to Junior High School ICT teacher.



What is a security ?

- ▶ No clear cut definition
- ▶ Security is a process, not an end state.
- ▶ Security is the process of maintaining an acceptable level of perceived risk.



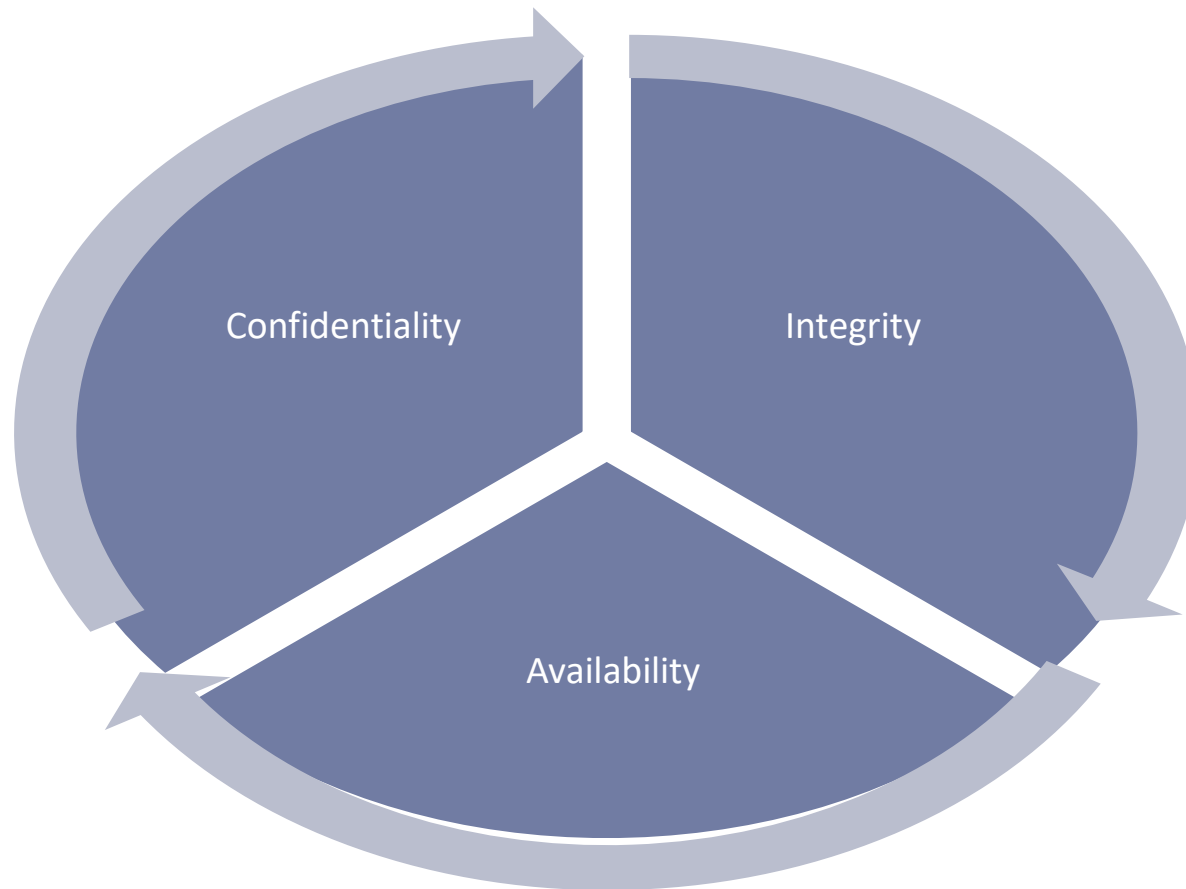
Security Management Concepts

- **What is Information Security?**
- Principles of Information Security

What is Information Security?

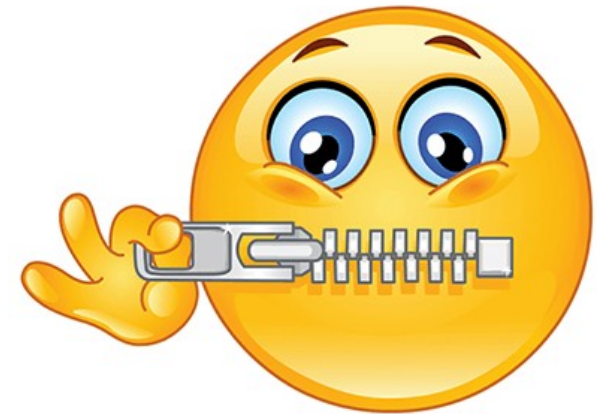
- Information Security (InfoSec) is the protection of:
 - Data
 - Devices
 - Networks
 - Software systems
- **Objectives**
 - Prevent unauthorized access
 - Maintain data privacy
 - Ensure business continuity

Security Features



Confidentiality

- ▶ Confidentiality is roughly equivalent to privacy. Measures undertaken to ensure confidentiality are designed to prevent sensitive information from reaching the wrong people, while making sure that the right people can in fact get it.
- ▶ Assurance that information is shared only among authorized persons or organizations.



Integrity

- ▶ Assurance that the information is authentic and complete. In information security, data integrity means maintaining and assuring the accuracy and consistency of data over its entire life-cycle.

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Availability

- ▶ Assurance that the systems responsible for delivering, storing and processing information are accessible when needed, by those who need them.
- ▶ Availability of information refers to ensuring that authorized parties are able to access the information when needed.

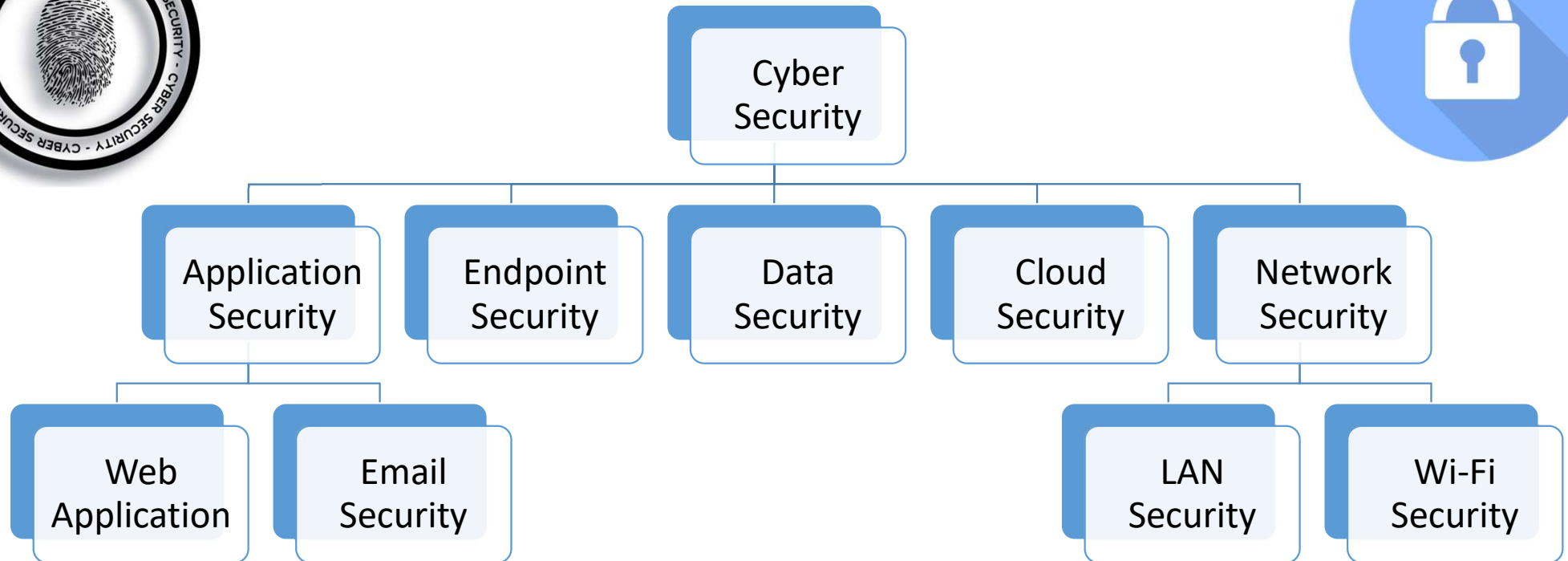


What is Cyber Security?

- Practice of protecting systems, networks, and programs from digital attacks.
- It involves a range of technologies, processes, and practices designed to safeguard computers, networks, programs, and data from unauthorized access, damage, or attacks.
- It encompasses everything from antivirus software and firewalls to encryption and multifactor authentication, creating a multi-layered defense against potential threats.



Cyber Security



Required Skills for Ethical Hacking

- Programming Languages: C and Python
- Linux
- Networking Skills
- Scripting
- Wireshark
- Cryptography Skills
- Experiment a Lot

Hacking Terms



Threat

- “A potential violation of security”
- Threat - A negative effect or undesired event. Threat is a potential occurrence described as an effect that might damage or compromise the asset.
- Threats can be classified as :
 - Threat of exposure, which is against on confidentiality of the system.
 - Threat of modification - which is against on integrity
 - Denial of Service - which is against on availability

Threats to Assets



Hardware - Can be stolen, broken or disabled

Software - Can be deleted, copied or modified to do unintended tasks

Data - Can be deleted, read, or modified

Communication lines - Can be destroyed, messages can be read, delayed, modified and fabricated



Vulnerability and Attack

- Vulnerability - Vulnerability of a computer system is a weak point which makes such a threat possible. A feature or bug in the system or program in which enables an attacker to bypass security measures.
- Attack - An attack is action performed by an attacker which the aim of finding and exploiting some of the vulnerabilities. Therefore an attack is a threat that is carried out.

Hackers Class



- Black Hat



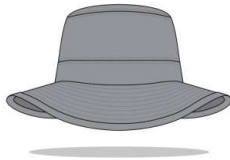
- “A person with extraordinary computing skills involved in malicious or destructive activities“

- White Hat



- “Person possessing hackers skill using them for defensive purpose aka security analyst”

- Gray Hat



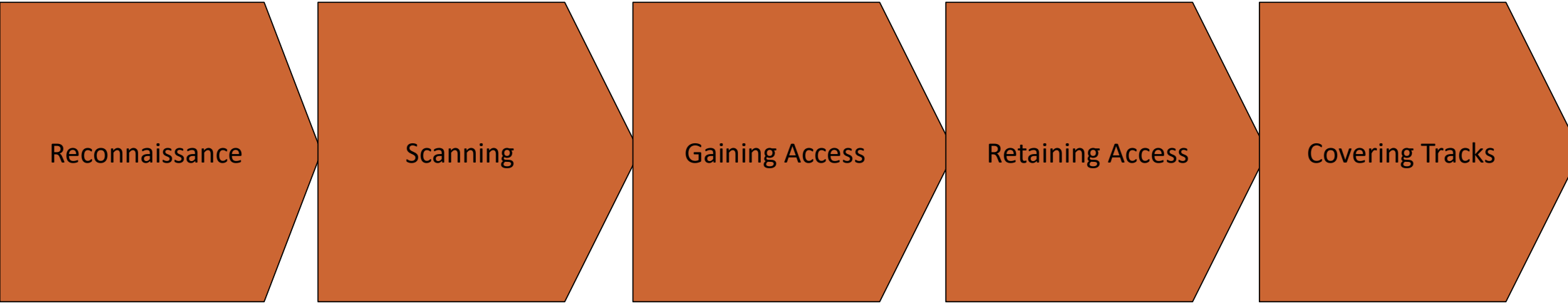
- “Person who plays a role of black hat and white hat at various times”

- Suicide Hackers

- “A person committed to bring down critical infrastructure without worrying to face punishments”

Basic Steps Of Hacking

- Reconnaissance
- Scanning
- Gaining Access
- Retaining Access
- Covering Tracks



Reconnaissance

Scanning

Gaining Access

Retaining Access

Covering Tracks

Reconnaissance

The first phase in the hacking approach, called “Reconnaissance also called as Information Gathering/Footprinting,” is all about acquiring data.

The majority of organizations communicate a great deal of information and data via a variety of channels, including their websites, social media accounts, help desks, and even workers.

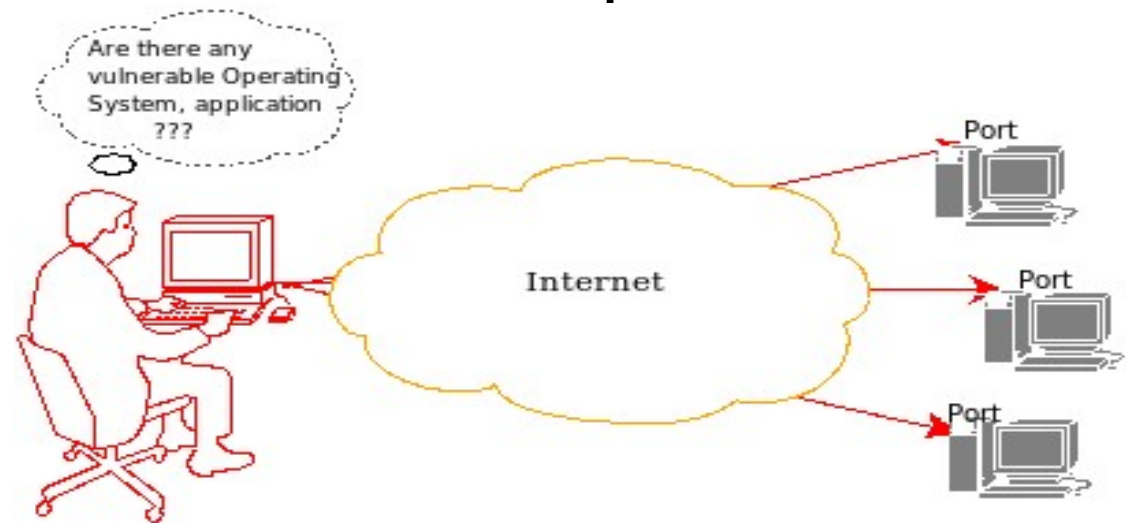


Scanning

This is the second phase of ethical hacking, Scanning refers to the pre-attack phase when the hacker scans the network for specific information on the basis of information gathered during reconnaissance. This phase also includes sniffing of network traffic if required.

Scanning includes

- Port scanners
- Vulnerability scanners
- etc.



Gaining Access

Gaining Access refers to the penetration phase. The hacker exploits the vulnerability in the target of evaluation

- Gaining of access can be achieved by
 - Buffer overflows
 - Denial of services
 - Session hijacking
 - Password cracking



Retaining Access

Retaining Access refers to the phase when the hacker tries to retain the ownership of the system

The hacker has compromised the system

Hackers may harden the system from other hackers as well

Hackers can upload, download or manipulate data, applications or configurations on the owned system

Covering Tracks

Covering Track refers to the activities that the hacker undertakes to hide his misdeed

Removing evidence of hacking or avoiding legal action



Ethical Hacking Teams

- Red Team
 - **Red Teams** are offensive security focused. They simulate how a possible attacker would attack cybersecurity defenses.
- Blue Team
 - **Blue Teams** are defense focused. They architect and maintain the protective internal cybersecurity infrastructure.
- Purple Team
 - **Purple Teams - Blue Red Teams** - are both offensively and defensively minded and were designed to ensure holistic and synergistic operations and information exchange between attackers and company defenders. Typically purple teams aren't really a team at all, rather a collaborative agreement between red and blue teams.

Red Team

- The red team takes on the role of an attacker, employing advanced attack methods to discover and try to exploit any possible vulnerabilities in the organization's cyber defenses.
- These offensive teams are usually made up of highly skilled security experts or freelance ethical hackers who concentrate on penetration testing by mimicking actual attack strategies.
- **Red team activities**
 - Penetration testing
 - Wireless access
 - Social engineering
 - Intercepting communication

Red Team Certification

- Certified Ethical Hacker (CEH)
- Licensed Penetration Tester (LPT) Master
- CompTIA PenTest+
- GIAC Penetration Tester (GPEN)
- GIAC Exploit Researcher and Advanced Penetration Tester (GXPN)
- Offensive Security Certified Professional (OSCP)
- Certified Red Team Operations Professional (CRTOP)

Blue Team

- When the red team is on offensive, the blue team is defending.
- This team often consists of incident response experts that advise the IT security team on how to strengthen their defences against increasingly sophisticated cyberthreats.
- The internal network's risk management is then the responsibility of the IT security team.
- **Blue team activities**
 - DNS audits
 - Installing and configuring firewalls and endpoint security software
 - Monitoring network activity
 - Deploying IDS and IPS software as a detective and preventive security control.
 - Implementing SIEM solutions to log and ingest network activity.

Blue Team Certifications

- Certified Information Systems Security Professional (CISSP)
- Certified Information Systems Auditor (CISA)
- CompTIA Security+
- GIAC Security Essentials Certification (GSEC)
- GIAC Certified Incident Handler (GCIH)
- Systems Security Certified Practitioner (SSCP)
- CompTIA Advanced Security Practitioner (CASP+)

Introduction to Kali Linux

- Kali Linux is an open-source, Debian-based Linux distribution geared towards various information security tasks, such as Penetration Testing, Security Research, Computer Forensics and Reverse Engineering.
- <https://www.kali.org/downloads/>



Installation of KALI on VM

← → ↻ 🏠 🔒 https://www.kali.org/get-kali//kali-platforms 📄 ☆ 📧 ⬇️ 🖨️

KALI GET KALI BLOG DOCUMENTATION ▾ COMMUNITY ▾ COURSES ▾ DEVELOPERS ▾ ABOUT ▾

• Demo

Choose **your** Platform

LIGHT ☒ DARK



Installer Images

- ✓ Direct access to hardware
- ✓ Customized Kali kernel
- ✓ No overhead

Single or multiple boot Kali, giving you complete control over the hardware access (perfect for in-built Wi-Fi and GPU), enabling the best performance.

💡 Recommended



Virtual Machines

- ✓ Snapshots functionary
- ✓ Isolated environment
- ✓ Customized Kali kernel
- ✗ Limited direct access to hardware
- ✗ Higher system requirements

VMware & VirtualBox pre-built images. Allowing for a Kali install without altering the host OS with additional features such as snapshots. Vagrant images for quick spin-up also available.

💡 Recommended

References

- ▶ <https://www.pearsonitcertification.com/articles/article.aspx?p=3129461&seqNum=3>
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